GOVERNMENT OF WESTERN AUSTRALIA

GSWA Geochronology Record

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Department of Energy, Mines, Industry Regulation and Safety

242683.1.pdf *Geochronology Record 1986*

242683.1: metarhyolite, Golden Stream open pit

(Meekatharra Formation, Polelle Group, Murchison Supergroup; Youanmi Terrane, Yilgarn Craton)

MTD Wingate, Y Lu, IOH Fielding and TJ Ivanic

Location and sampling

YALGOO SH50-02, MELLENBYE 2140 Zone 50, MGA 440729E 6831071N
Warox Site TJIGUL190012 Sampled on 6 May 2019

This sample was collected from an outcrop southwest of the Cagacaroon Hills, about 5.5 km south of Armstrong Well, 2.2 km north of Gearless Well, and 0.2 km northeast of the Golden Stream open pit.

Geological context

The unit sampled is a metarhyolite (Fig. 1) of the Meekatharra Formation of the Polelle Group. The 2800–2760 Ma Meekatharra Formation is about 2.5 km thick and composed mainly of weakly metamorphosed basalt and komatiitic basalt, with thin interflow felsic metavolcanic and metasedimentary rocks (Van Kranendonk et al., 2013, and references therein). The Meekatharra Formation conformably overlies the Coodardy Pool Formation at the base of the 2800–2735 Ma Polelle Group and is conformably overlain by the Greensleeves or Wilgie Mia Formations. The present sample was collected near the northwestern margin of the Gullewa greenstone belt, west of the Yalgoo Dome (Zibra et al., 2018), from a north-trending unit of laminated metarhyolite, about 1 m thick, between metabasalt to the west and metavolcaniclastic rocks to the east. About 3.4 km to the south-southwest, a hydrothermally altered volcaniclastic breccia yielded zircon age components at c. 2967, 2757, 2731, 2700 and 2671 Ma (GSWA 211120, Wingate et al., 2021), although the youngest age component is based on dates from high-U zircons and now thought likely to reflect Pb loss associated with hydrothermal alteration. A hornblende metamonzonite sampled about 4.2 km to the west yielded an igneous crystallization age of 2754 ± 4 Ma (GSWA 242695, Wingate et al., 2022).

Figure 1. Hand specimen image of sample 242683: metarhyolite, Golden Stream open pit

Petrographic description

The sample is a weakly laminated, fine- to medium-grained metarhyolite (Fig. 1), consisting of about 80% microcrystalline groundmass, 15–20% biotite, and 2–3% iron oxide minerals. The groundmass is assumed to consist mainly of quartz, because no feldspar was identified. The groundmass has an average grain size of about 20 µm, although a minor amount of dispersed, subangular quartz grains are up to 0.1 mm in size. Subhedral laths of brown biotite are up to 0.1 mm in size,

although most are 0.05 mm or smaller. Slightly larger laths are concentrated in decussate aggregates that form equant patches up to 0.5 mm diameter and minor lenses up to 1 mm long. The lamination is poorly defined by patches and lenses, up to 1 mm long, of very fine-grained orange–brown iron oxide minerals, and to a much lesser extent by slight variations in grain size. The sample is not obviously deformed. Biotite is interpreted as metamorphic, and the lack of preferred orientation suggests it may be the product of contact metamorphism at biotite grade.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown or opaque and subhedral to euhedral. The crystals are up to 150 µm long, and equant to slightly elongate, with aspect ratios up to 3:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous. A CL image of representative zircons is shown in Figure 2.

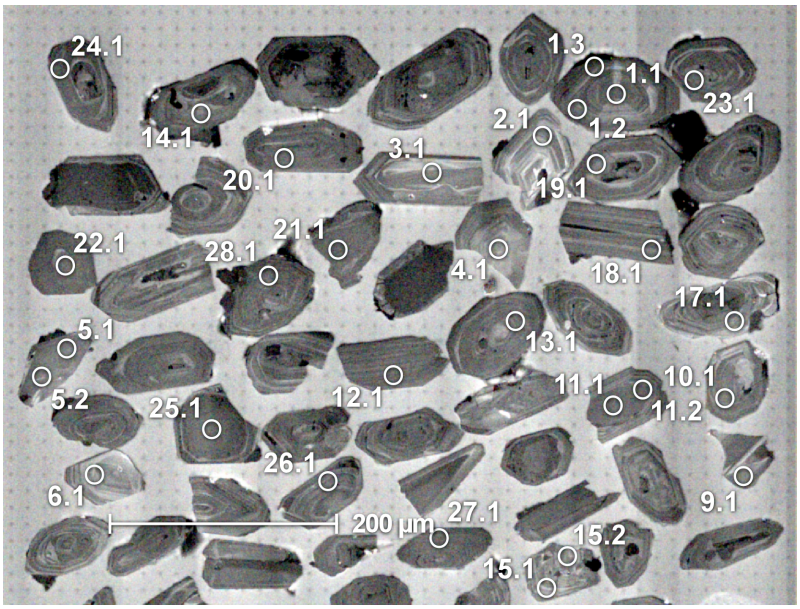

Figure 2. Cathodoluminescence image of representative zircons from sample 242683: metarhyolite, Golden Stream open pit. Numbered circles indicate the approximate locations of analysis sites

Analytical details

This sample was analysed over three sessions, on 20–21 March, 26–27 March, and 14–15 May, 2020, using SHRIMP-B. Analyses 1.1 to 14.1 (spot numbers 1–14) were obtained during the first session, together with 12 analyses of the M257 standard, of which 11 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 1.2% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.42% (1σ). Analyses 15.1 to 22.1 (spot numbers 15–26) were obtained during the second session, together with 11 analyses of the M257 standard, of which eight analyses indicated an external spot-to-spot (reproducibility) uncertainty of 2.33% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.99% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the second session was corrected by reference to the OGC1 standard; measured ratios were decreased by 0.44%. Analyses 1.3 to 28.1 (spot numbers 27–34) were obtained during the third session, together with 10 analyses of the M257 standard, which indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.20% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the third session was corrected by reference to the OGC1 standard; measured ratios were decreased by 0.54%. Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1 (provided in the Links section). Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Thirty-four analyses were obtained from 28 zircons. Results are listed in Table 1 and shown in concordia diagrams (Figs 3 and 4).

Interpretation

The analyses are concordant to strongly discordant (Fig. 3). Eleven analyses are >5% discordant. The dates obtained from these 11 analyses (Group D; Table 1) are unreliable and are considered not to be geologically significant. The remaining 23 analyses can be divided into two groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises seven analyses of five zircons (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2793 ± 8 Ma (MSWD = 1.3).

Group X comprises 16 analyses of 15 zircons (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 3031–2922 Ma. Fourteen analyses of 13 zircons in Group X yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2947 ± 5 Ma (MSWD = 1.2).

The date of 2793 ± 8 Ma for the seven analyses in Group I is interpreted as the magmatic crystallization age. The dates of 3031–2922 Ma for the 16 analyses in Group X are interpreted as the ages of inherited zircons. The date of 2947 ± 5 Ma for 14 analyses in Group X is interpreted as the age of an inherited component.

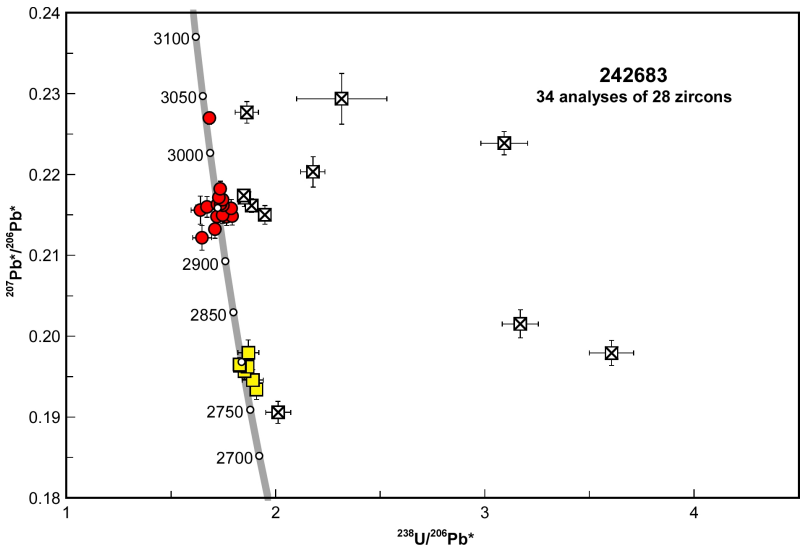

Figure 3. **U–Pb analytical data for zircons from sample 242683: metarhyolite, Golden Stream open pit. Yellow squares indicate Group I (magmatic zircons); red circles indicate Group X (xenocrystic zircons); crossed squares indicate Group D (discordance >5%)**

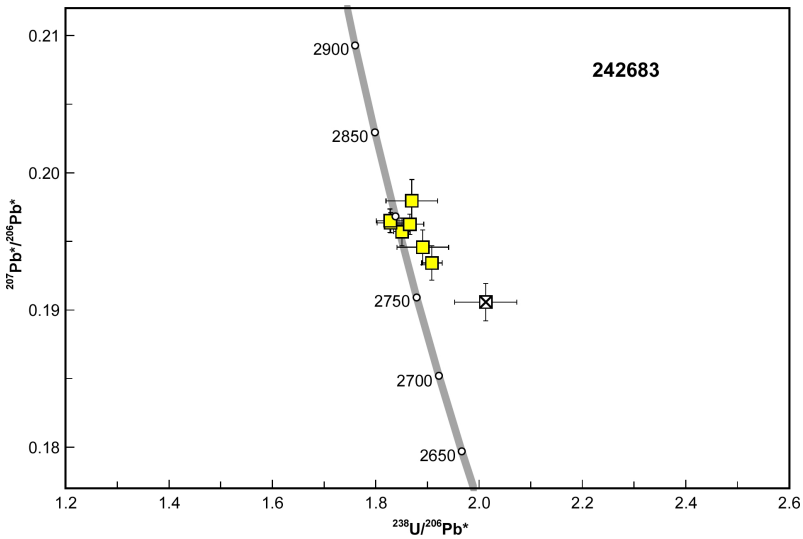

Figure 4. **Expanded view of U–Pb analytical data for zircons from sample 242683: metarhyolite, Golden Stream open pit. Symbols as in Figure 3**

Links

Ion microprobe analytical results for zircons from sample 242683: metarhyolite, Golden Stream open pit: 242683.1.Table1.xls
SHRIMP spot images: 242683_spots.pdf

Introduction to geochronology information: Intro_2024.pdf

References

Stacey, JS and Kramers, JD 1975, Approximation of terrestrial lead isotope evolution by a two-stage model: Earth and Planetary Science Letters, v. 26, p. 207–221.

Van Kranendonk, MJ, Ivanic, TJ, Wingate, MTD, Kirkland, CL and Wyche, S 2013, Long-lived, autochthonous development of the Archean Murchison Domain, and implications for Yilgarn Craton tectonics: Precambrian Research, v. 229, p. 49–92.

Wingate, MTD, Fielding, IOH, Lu, Y and Ivanic, TJ 2022, 242695: hornblende metamonzonite, Olivers Well; Geochronology Record 872: Geological Survey of Western Australia, 4p., <www.dmirs.wa.gov.au/geochron>.

Wingate, MTD, Lu, Y, Fielding, IOH and Ivanic, TJ 2021, 211120: altered breccia, Deflector prospect; Geochronology Record 1823: Geological Survey of Western Australia, <www.dmirs.wa.gov.au/geochron>. View Reference online

Zibra, I, Peternell, M, Schiller, M, Wingate, MTD, Lu, YJ and Clos, F 2018, Tectono-magmatic evolution of the Neoarchean Yalgoo dome (Yilgarn Craton): diapirism in a pre-orogenic setting: Geological Survey of Western Australia, Report 176, 43p.

Recommended reference for this publication

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Grid references in this publication refer to the Geocentric Datum of Australia 2020 (GDA20). All locations are quoted to at least the nearest 100 m.

Capitalized names in text refer to standard 1:100 000 map sheets, unless otherwise indicated.

WAROX is GSWA's field observation and sample database. WAROX site IDs have the format 'ABCXXXnnnnnnSS', where ABC = geologist username, XXX = project or map code, nnnnnn = 6-digit site number, and SS = optional alphabetic suffix (maximum 2 characters).

Isotope and element analyses are routinely conducted using the GeoHistory laser ablation ICP-MS and Sensitive High-Resolution Ion Microprobe (SHRIMP) facilities at the John de Laeter Centre (JdLC), Curtin University, with the financial support of the Australian Research Council and AuScope National Collaborative Research Infrastructure Strategy (NCRIS), and the Microscopy Australia laser ablation ICP-MS facility at the Centre for Microscopy, Characterisation & Analysis (CMCA), The University of Western Australia (UWA). The TESCAN Integrated Mineral Analyser (TIMA) instrument was funded by a grant from the Australian Research Council (LE140100150) and is operated by the JdLC with the support of the Geological Survey of Western Australia, UWA and Murdoch University. Mineral analyses are routinely obtained using the electron probe microanalyser (EPMA) facilities at the CMCA at UWA, at Adelaide Microscopy at the University of Adelaide and the Central Science Laboratory at the University of Tasmania.

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